

Spatial Statistics - Research

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1 A General Periodic Fourier–State-Space Spatio–Temporal Model

1.1 Model setup

We consider a real-valued Gaussian spatio–temporal process

$$\{X_t(x) : x \in [0, 2\pi], t \in \mathbb{Z}\},$$

defined on a one-dimensional spatial domain with periodic boundary conditions (a circle) and discrete time. For each time t , the spatial field $X_t(\cdot)$ is represented by a truncated Fourier series

$$X_t(x) = a_0(t) + \sum_{j=1}^J \left[a_j(t) \cos(jx) + b_j(t) \sin(jx) \right], \quad x \in [0, 2\pi], \quad (1)$$

where J is a truncation level and

$$a_j(t), b_j(t), \quad j = 0, \dots, J, t \in \mathbb{Z},$$

are jointly Gaussian random coefficients.

For $j \geq 1$, the pair $(a_j(t), b_j(t))$ corresponds to the j th spatial frequency (wavenumber). Low j terms encode large-scale, slowly varying spatial structure, while high j terms encode fine-scale spatial detail.

1.2 Spatial margin: Matérn-like variances

The spatial covariance at each fixed time is controlled by a Matérn-like spectral density in the Fourier domain. For $j = 0, \dots, J$ we specify

$$\sigma_j^2 \propto (\kappa^2 + j^2)^{-(\nu+1/2)}, \quad \kappa = \frac{\sqrt{2\nu}}{\ell}, \quad (2)$$

where

- $\nu > 0$ is the Matérn smoothness parameter,
- $\ell > 0$ is a spatial range parameter,
- $\kappa = \sqrt{2\nu}/\ell$ is an inverse scale parameter.

NOTE: This spectral form is appropriate for a *non-periodic* domain $\mathbb{R}$. On a periodic domain $[0, 2\pi]$, the correspondence between this discrete spectrum and the continuous Matérn model requires careful treatment.

Question: Should we (a) sample the continuous Matérn spectrum at integer frequencies j , or (b) explicitly periodize the Matérn covariance function and compute its Fourier coefficients?

For simulation we normalize these variances so that the total marginal variance of $X_t(x)$ is σ_{tot}^2 . Defining

$$\tilde{\sigma}_j^2 = (\kappa^2 + j^2)^{-(\nu+1/2)},$$

we set

$$\sigma_j^2 = \sigma_{\text{tot}}^2 \cdot \frac{\tilde{\sigma}_j^2}{\sum_{k=0}^J \tilde{\sigma}_k^2}, \quad j = 0, \dots, J. \quad (3)$$

1.3 Temporal dynamics: block-diagonal AR(1) with rotation and correlated innovations

We collect the Fourier coefficients into a state vector

$$\mathbf{Z}_t = (a_0(t), a_1(t), \dots, a_J(t), b_1(t), \dots, b_J(t))^\top \in \mathbb{R}^{2J+1}. \quad (4)$$

The temporal evolution is governed by a *block-diagonal* linear Gaussian state-space model. For the constant mode $j = 0$, we have a scalar AR(1):

$$a_0(t+1) = \rho_0 a_0(t) + \eta_{0,t}, \quad \eta_{0,t} \sim \mathcal{N}(0, \sigma_0^2(1 - \rho_0^2)), \quad (5)$$

with $|\rho_0| < 1$ ensuring stationarity.

For each mode $j \geq 1$, the cosine and sine coefficients evolve jointly according to a 2×2 block AR(1):

$$\begin{pmatrix} a_j(t+1) \\ b_j(t+1) \end{pmatrix} = M_j \begin{pmatrix} a_j(t) \\ b_j(t) \end{pmatrix} + \begin{pmatrix} \eta_{a,j,t} \\ \eta_{b,j,t} \end{pmatrix}, \quad (6)$$

where the 2×2 transition matrix is

$$M_j = \rho_j \begin{pmatrix} \cos \theta_j & -\sin \theta_j \\ \sin \theta_j & \cos \theta_j \end{pmatrix}, \quad (7)$$

with $\rho_j \in (0, 1)$ (amplitude persistence) and $\theta_j \in \mathbb{R}$ (rotation angle).

The 2×2 innovation covariance for mode j is

$$\begin{pmatrix} \eta_{a,j,t} \\ \eta_{b,j,t} \end{pmatrix} \sim \mathcal{N}(\mathbf{0}, \Sigma_{\varepsilon,j}), \quad \Sigma_{\varepsilon,j} = \sigma_j^2(1 - \rho_j^2) \begin{pmatrix} 1 & \phi_j \\ \phi_j & 1 \end{pmatrix}, \quad (8)$$

where $\phi_j \in (-1, 1)$ is the *within-frequency correlation parameter* between the cosine and sine innovations.

WARNING — EMPIRICAL VALIDATION ISSUE: The Lyapunov equation $\Sigma_j = M_j \Sigma_j M_j^\top + \Sigma_{\varepsilon,j}$ with target stationary covariance $\Sigma_j = \sigma_j^2 \begin{pmatrix} 1 & \phi_j \\ \phi_j & 1 \end{pmatrix}$ does yield (8) as shown in Section 1.4. However, empirical validation showed:

- Empirical $\text{Cor}(a_1, b_1) = 0.208$, theoretical $\phi_1 = 0.696$ (70% error)
- Phase drift empirically ≈ 0 even when $\theta_j \neq 0$ was set
- ACF of amplitude decays faster than ρ_j^k (up to 57% error for high j)

This suggests either (a) an implementation error in the R code, or (b) a misunderstanding of what quantity should match theory when rotation is present.

The innovations are independent across different frequencies j and across time t .

Physical interpretation

Rotation parameter θ_j : When $\theta_j = 0$, the transition matrix $M_j = \rho_j I_2$, so cosine and sine evolve independently. When $\theta_j \neq 0$, we have temporal mixing:

$$a_j(t+1) = \rho_j [\cos \theta_j \cdot a_j(t) - \sin \theta_j \cdot b_j(t)] + \eta_{a,j,t},$$

inducing *phase rotation*: spatial patterns propagate around the circle over time. The eigenvalues of M_j are $\rho_j e^{\pm i\theta_j}$, so stationarity holds for $|\rho_j| < 1$ regardless of θ_j .

Question: Does phase rotation θ_j have a physical interpretation in terms of wave propagation speed on the circle? If $\theta_j = c \cdot j$ for constant c , does this correspond to a traveling wave with speed c ?

Innovation correlation parameter ϕ_j : When $\phi_j = 0$, the innovations $\eta_{a,j,t}$ and $\eta_{b,j,t}$ are independent. When $\phi_j \neq 0$, the innovations are correlated: $\text{Cov}(\eta_{a,j,t}, \eta_{b,j,t}) = \sigma_j^2(1 - \rho_j^2)\phi_j$. At stationarity, this targets $\text{Cov}(a_j(t), b_j(t)) = \sigma_j^2 \phi_j$ via the Lyapunov equation (Section 1.4).

Typical parameterization: $\phi_j = \phi_0/(1+j)$ or $\phi_j = \phi_0 e^{-\gamma j}$.

Combined effect: $\theta_j \neq 0$ creates *temporal cross-dependence*; $\phi_j \neq 0$ creates *contemporaneous correlation*. Together they allow rich within-frequency dynamics while maintaining the block-diagonal structure.

Key empirical finding: Simulated $\text{Cor}(a_j, b_j)$ showed 70% error for $j = 1$ and wild scatter for other modes, suggesting either incorrect initial condition draws or a numerical issue in the multivariate normal generation.

1.4 Stationary covariance of $(a_j(t), b_j(t))$ via Lyapunov equation

At stationarity, the 2×2 covariance matrix

$$\Sigma_j = \text{Cov} \begin{pmatrix} a_j(t) \\ b_j(t) \end{pmatrix}$$

satisfies the discrete-time Lyapunov equation

$$\Sigma_j = M_j \Sigma_j M_j^\top + \Sigma_{\varepsilon,j}. \quad (9)$$

Corrected derivation: For the rotation-scaling matrix (7), we have

$$M_j M_j^\top = \rho_j^2 I_2,$$

so (9) becomes

$$\Sigma_j = \rho_j^2 \Sigma_j + \Sigma_{\varepsilon,j} \implies \Sigma_{\varepsilon,j} = (1 - \rho_j^2) \Sigma_j.$$

If we target the stationary covariance

$$\Sigma_j = \sigma_j^2 \begin{pmatrix} 1 & \phi_j \\ \phi_j & 1 \end{pmatrix},$$

then the innovation covariance must be

$$\Sigma_{\varepsilon,j} = \sigma_j^2 (1 - \rho_j^2) \begin{pmatrix} 1 & \phi_j \\ \phi_j & 1 \end{pmatrix}.$$

This formula is *independent of* θ_j because $M_j M_j^\top = \rho_j^2 I_2$ for any rotation angle. Therefore equation (8) is theoretically correct, and the empirical mismatch must arise from implementation or interpretation errors.

Advisor questions:

1. The empirical ACF of amplitude $R_j(t) = \sqrt{a_j(t)^2 + b_j(t)^2}$ decays much faster than ρ_j^k (up to 57% error for high j). Should the theoretical ACF be computed for the *amplitude* instead of the coefficients?
2. The empirical correlation $\text{Cor}(a_j(t), b_j(t))$ showed wild scatter instead of matching $\phi_j \approx 0.65$ – 0.70 . Is this a finite-sample effect, or is there a bias in how the stationary initial condition is drawn?
3. The spatial covariance at large lags showed an “upturn” (U-shape) instead of monotonic decay. Is this the “ridge” artifact from nonseparable models warned about in Stein (2005)? Does this indicate we need different temporal structure?

Therefore, at stationarity:

$$\text{Var}(a_j(t)) = \sigma_j^2, \quad (10)$$

$$\text{Var}(b_j(t)) = \sigma_j^2, \quad (11)$$

$$\text{Cov}(a_j(t), b_j(t)) = \sigma_j^2 \phi_j. \quad (12)$$

The stationary covariance depends on ϕ_j but *not* on θ_j . The rotation angle affects temporal evolution but not marginal variances or contemporaneous correlation.

1.5 Nonseparability via frequency-dependent persistence

The key nonseparable feature arises from allowing ρ_j to depend on spatial frequency j . We propose an *algebraic (power-law) decay* parameterization

$$\rho_j = \frac{1}{(1 + \lambda j^\alpha)^\beta}, \quad j = 1, \dots, J, \quad (13)$$

with $\rho_0 \in (0, 1)$ set separately, and parameters $\lambda > 0$, $\alpha > 0$, $\beta > 0$.

Under this structure, low-frequency modes have $\rho_j \approx 1$ (slow temporal decay), while high-frequency modes have smaller ρ_j (fast decay). The effective spatial correlation structure changes with time lag: at short lags both low and high frequencies contribute, but at long lags only low frequencies remain. This space–time covariance cannot be written as $C_{\text{space}}(h)C_{\text{time}}(k)$ and is therefore nonseparable.

1.6 Space–time covariance

Under the block-diagonal structure, the space–time covariance

$$C(h, k) = \mathbb{E}[X_t(x)X_{t+k}(x+h)]$$

can be computed when different modes are uncorrelated across frequencies. For $\theta_j = 0$ (no rotation), we have

$$\mathbb{E}[a_j(t)a_j(t+k)] = \mathbb{E}[b_j(t)b_j(t+k)] = \sigma_j^2 \rho_j^{|k|},$$

yielding

$$C(h, k) = \sum_{j=0}^J \sigma_j^2 \rho_j^{|k|} \cos(jh). \quad (14)$$

Derivation assumes $\theta_j = 0$: When M_j includes rotation, the lag- k covariance involves M_j^k :

$$M_j^k = \rho_j^k \begin{pmatrix} \cos(k\theta_j) & -\sin(k\theta_j) \\ \sin(k\theta_j) & \cos(k\theta_j) \end{pmatrix},$$

so

$$\mathbb{E}[a_j(t)a_j(t+k)] = \sigma_j^2 \rho_j^k \cos(k\theta_j) + \text{cross-terms involving } \phi_j.$$

The spatial covariance (14) is therefore correct *only when $\theta_j = 0$* or when cross-terms cancel in the Fourier reconstruction. **please verify when rotation is present.**

1.7 Parameter overview

The full model is specified by the following parameters, organized by their role in the model structure.

Fourier truncation

- $J \in \mathbb{Z}_+$: highest Fourier mode (truncation level). Controls the finest spatial scale represented. Larger J allows finer spatial detail; in practice J is chosen so that the total variance in modes $j > J$ is negligible.

Spatial margin parameters (Matérn-like spectrum)

- $\nu > 0$: Matérn smoothness parameter. Larger ν gives smoother spatial fields. Sample paths are $\lceil \nu \rceil - 1$ times mean-square differentiable.
- $\ell > 0$: spatial range parameter. Larger ℓ increases the spatial correlation length by shifting spectral mass to lower frequencies.
- $\sigma_{\text{tot}}^2 > 0$: total marginal variance of the field $X_t(x)$. Overall scale parameter.
- $\kappa = \sqrt{2\nu}/\ell$: (derived) inverse range parameter used in the spectral density (2).

The mode variances σ_j^2 for $j = 0, \dots, J$ are derived from $(\nu, \ell, \sigma_{\text{tot}}^2)$ via (2) and (3).

Temporal persistence parameters

- $\rho_0 \in (0, 1)$: AR(1) persistence for the constant mode $a_0(t)$. Set separately to avoid nonstationarity (the formula (13) would give $\rho_0 = 1$ for $j = 0$).
- $\lambda > 0$: controls overall temporal decay strength across frequencies in (13).
- $\alpha > 0$: controls how strongly persistence changes with frequency j in (13).
- $\beta > 0$: decay exponent in (13). Smaller β gives slower (longer-memory) decay.

The mode-specific persistence parameters ρ_j for $j = 1, \dots, J$ are derived from (λ, α, β) via (13).

Phase rotation parameters

- $\theta_j \in \mathbb{R}$ for $j = 1, \dots, J$: rotation angle for mode j in the 2×2 transition matrix M_j (equation (7)).
- When $\theta_j = 0$ for all j , the model reduces to independent cosine/sine evolution (no temporal mixing).
- Typical parameterization: $\theta_j = \theta_0/j$ or $\theta_j = \theta_0/(1+j)$ for a scalar hyperparameter $\theta_0 \in \mathbb{R}$, so that lower frequencies rotate more than higher frequencies.

Empirical finding: Phase drift validation showed mean $\Delta\text{phase} \approx 0$ even when θ_j was set to nonzero values. This suggests either (a) rotation is not being applied correctly in simulation, or (b) the phase of amplitude $R_j(t) = \sqrt{a_j^2 + b_j^2}$ does not drift at rate θ_j as expected. **What is the theoretical expected phase drift rate when rotation is present?**

Innovation correlation parameters

- $\phi_j \in (-1, 1)$ for $j = 1, \dots, J$: within-frequency correlation between innovations $\eta_{a,j,t}$ and $\eta_{b,j,t}$ (equation (8)).
- When $\phi_j = 0$ for all j , the innovations are independent (cosine and sine uncorrelated at the same time).
- When $\phi_j \neq 0$, the model induces contemporaneous correlation $\text{Cov}(a_j(t), b_j(t)) = \sigma_j^2 \phi_j$ at stationarity.
- Typical parameterization: $\phi_j = \phi_0/(1+j)$ or constant $\phi_j = \phi_0$ for all j , where $\phi_0 \in (-1, 1)$ is a scalar hyperparameter.

Empirical finding: Simulated $\text{Cor}(a_j, b_j)$ showed 70% error for $j = 1$ (empirical 0.208 vs. theoretical 0.696) and high scatter for other modes. This suggests either (a) the stationary initial condition is drawn incorrectly, or (b) there is a numerical issue in the multivariate normal generation. The Lyapunov theory is correct (Section 1.4), so the issue is implementation-related.

Time horizon

- $T \in \mathbb{Z}_+$: number of time points to simulate or observe. Not a model parameter but a data specification.

Parameter count summary

For a model with Fourier truncation J and the simplest scalar hyperparameter choices, the total number of free parameters is:

- **Minimal model** (independent modes, no rotation, no innovation correlation): $J = 50$ modes

$$\text{Parameters: } \nu, \ell, \sigma_{\text{tot}}^2, \rho_0, \lambda, \alpha, \beta \quad \Rightarrow \quad 7 \text{ parameters total.}$$

- **With rotation** (add θ_0 for $\theta_j = \theta_0/j$):

$$\text{Parameters: } 7 + 1 = 8.$$

- **With rotation and innovation correlation** (add θ_0 and ϕ_0 for $\phi_j = \phi_0/(1+j)$):

$$\text{Parameters: } 7 + 1 + 1 = 9.$$

- **Fully flexible** (separate θ_j and ϕ_j for each mode $j = 1, \dots, J$):

$$\text{Parameters: } 7 + J + J = 7 + 2J.$$

For $J = 50$, this is 107 parameters—likely overparameterized for most applications.

Question: For inference (e.g., Whittle likelihood in Fourier space), which parameterization strikes the best balance between flexibility and identifiability? Should we use mode-specific $\{\rho_j, \theta_j, \phi_j\}$ as free parameters, or constrain them via smooth functions of j with scalar hyperparameters?

By setting $\theta_j = 0$ and $\phi_j = 0$ for all j , the model reduces to the simple independent-mode AR(1) with uncorrelated cosine and sine components (7 parameters). Nonzero θ_j and ϕ_j introduce richer within-frequency dynamics: temporal mixing and phase rotation (θ_j), and instantaneous correlation (ϕ_j).

Open Questions

1. **Periodic Matérn spectrum:** Should we sample the continuous Matérn spectral density at integer frequencies j , or explicitly periodize the Matérn covariance and compute its Fourier series?
2. **Rotation and ACF:** When M_j includes rotation ($\theta_j \neq 0$), what is the theoretical ACF of amplitude $R_j(t) = \sqrt{a_j^2 + b_j^2}$? Does it match ρ_j^k , or does rotation change the decay rate?
3. **Innovation correlation identifiability:** The empirical $\text{Cor}(a_j, b_j)$ showed high variance. Is ϕ_j identifiable from data, or does it trade off with θ_j ?
4. **Phase drift interpretation:** If $\theta_j = c \cdot j$, does this correspond to a traveling wave with speed c ? Physical meaning of frequency-dependent rotation?
5. **Cross-frequency dependence:** Current model has block-diagonal dynamics. Should we add off-block-diagonal elements in M or Σ_ϵ for cross-frequency coupling? What structure preserves computational efficiency?
6. **Parameter parsimony for inference:** For Whittle likelihood estimation, should we use mode-specific $\{\rho_j, \theta_j, \phi_j\}$ as free parameters, or constrain via smooth functions with scalar hyperparameters (e.g., $\theta_j = \theta_0/j$)? What is the trade-off between flexibility and identifiability?

2 Next steps

Write the Covariance in closed form as then it nice to simulate with fft due to circulant structure.

If we truncate to J it depends on how dense the items are

Want $Z(x) = \sum a_j(t) \cos(jx) + b_j(t) \sin(jx)$ to be a Gaussian process with covariance $C(h, k)$.

Consider Aliasing $Z(k/n) = \sum a_j(t) \cos(2\pi jk/n) + b_j(t) \sin(2\pi jk/n)$

Now simulat $A'_j = \sum_p A_{a_j+pn}$ and $B'_j = \sum_p B_{b_j+pn}$

and $\text{Var}(Z(k/n)) = \sum_{p=-\infty}^{\infty} \sigma_{j+pn}^2 = \sum_{p=-q}^q \text{exact} + \sum_{p=q+1}^{\infty} + \sum_{p=-\infty}^{-q-1}$ tail by approximation

Ask about the proof for the other of psd is $\int_d$
 Can use FFT for periodic stationary. Paper: Chan & Wood, Deitrich & Neuwsam
 maybe doing a smooth extension.
 EQ 9 in stien paper JASA 2005
 $Z(x, t + 1) = \int v(x - y)dy + \epsilon$